

MeitY Sovereign AI Initiative: Phase-1 Pilot Proposal & Exhaustive Standard Operating Procedure (SOP)

Document Ref: MEITY-PILOT-PROPOSAL-V2.0	Date: July 31, 2026
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1. EXECUTIVE SUMMARY & REGULATORY ALIGNMENT

This document presents the formal deployment operational plan for a **Phase-1 Pilot Sandbox Deployment** of **FlockML** across Ministry of Electronics and Information Technology (MeitY) and National Informatics Centre (NIC) infrastructure.

FlockML is an open-source, browser-native Edge AI compute protocol compiled in WebGPU and WebAssembly. It pools compute cycles across standard desktop PCs, smartphones, and edge servers to run and train machine learning models with **\$0.00 additional cloud infrastructure expense** and **zero raw data leakage**, fully aligned with the **Digital Personal Data Protection (DPDP) Act 2023** and **CERT-In Data Security Guidelines**.

2. PILOT KEY PERFORMANCE INDICATORS (KPIs) & ACCEPTANCE CRITERIA

PERFORMANCE PILLAR	TARGET METRIC	VERIFICATION PROTOCOL & THRESHOLD	PASS / FAIL ACCEPTANCE CRITERIA
Zero Cloud Compute Spend	100% Client Processing	Server billing audit (\$0.00 cloud GPU/CPU spend)	Passed if cloud GPU allocation = 0
Sovereign Privacy Guarantee	Zero Raw Data Leakage	Laplacian Differential Privacy ($\epsilon = 0.8$) packet dump audit	Passed if raw input data reconstruction = 0%
Bandwidth Optimization	80% Payload Compression	Int8 Quantized gradient payload size comparison	Passed if payload size $\leq$ 25% of Float32 baseline
Offline Network Resilience	Zero-Second Outage Recovery	WAN disconnect simulation during live inference loop	Passed if model continues via local OPFS SSD cache

3. EXHAUSTIVE 14-DAY DAY-BY-DAY TECHNICAL PLAN OF ACTION

Days 1–2: Coordinator VM Provisioning & Network Firewall Configuration

- **Day 1 (VM Provisioning):** Provision 1x basic Linux VM (Ubuntu 22.04 LTS, 2 vCPU, 4GB RAM, 20GB SSD) inside the NIC internal sandbox perimeter. Deploy the `flockml-coordinator` container via Docker/systemd.
- **Day 2 (Firewall & WebSocket TLS Setup):** Configure Nginx reverse proxy with TLS 1.3 encryption. Open inbound/outbound ports (HTTPS 443, WSS 8443) on internal subnet for WebSocket signaling.

```
# Systemd Coordinator Unit Setup (Day 1)
sudo systemctl enable flockml-coordinator --now
sudo nginx -t && sudo systemctl reload nginx
```

Days 3–4: Node Onboarding & WebGPU / WASM Discovery

- **Day 3 (Onboarding Portal Launch):** Launch node enrollment portal at `https://meity-pilot.gov.in`. Onboard 25–50 test workstations and mobile devices across Delhi and Kolkata nodes. No admin privileges or binary installation required.
- **Day 4 (Capability Discovery & OPFS Cache Audit):** Execute automated discovery handshake to verify client WebGPU WGSL shaders, WebAssembly SIMD fallback engines, and OPFS local SSD access handles.

Days 5–7: Federated Training & Dynamic Weight Synchronization

- **Day 5 (Model Initialization):** Initialize baseline model parameters (Emergency Triage / Indic Text Classifier) on the FedAvg Coordinator.
- **Day 6–7 (Distributed Micro-batch Execution):** Client devices process local forward/backward training passes during idle browser cycles, returning Int8 quantized gradients to the coordinator.

Days 8–10: Security Inspection & Cryptographic Privacy Audit

- **Day 8–9 (Wireshark Packet Capture):** Collaborate with NIC cybersecurity officers to perform full PCAP network packet captures on outbound client traffic. Verify that zero PII or raw text leaves client memory.
- **Day 10 (Differential Privacy Verification):** Verify that Laplacian noise ($\epsilon = 0.8$) and gradient norm clipping ($C = 1.0$) guarantee mathematical non-reversibility of model updates.

Days 11–14: Outage Stress Testing & Governance Handover

- **Day 11 (WAN Outage Simulation):** Disconnect WAN internet access to client nodes; verify that inference continues uninterrupted via local OPFS SSD weight storage.
- **Day 12 (Dynamic Churn Test):** Abruptly terminate 50% of active client browser tabs; confirm FedAvg Aggregator smoothly handles node dropouts without job failure.
- **Days 13–14 (Formal Sign-Off):** Compile benchmark report on VRAM savings, compute throughput, and privacy audit metrics for Director Ranjan and GM Compute.

4. ROLES & RESPONSIBILITIES MATRIX (RACI FRAMEWORK)

TASK / DELIVERABLE	SUPRATIM DHARA (FLOCKML)	MEITY TECHNICAL LIAISON	NIC SYSTEM ADMINISTRATOR
Coordinator VM Deployment	Accountable (A) / Responsible (R)	Consulted (C)	Support / Responsible (R)
Client Node Enrollment	Responsible (R)	Accountable (A)	Informed (I)
Network Packet Security Audit	Support (S)	Accountable (A) / Responsible (R)	Consulted (C)
Fault Tolerance Simulation	Responsible (R)	Accountable (A)	Support (S)

5. RESOURCE INFRASTRUCTURE & ADMINISTRATIVE PROVISIONS

Hardware & Server Provisions:

- **1x Server VM Instance:** Ubuntu 22.04 LTS (2 vCPU, 4GB RAM, 20GB SSD) inside MeitY/NIC internal network to host signaling coordinator.
- **25–50 Client Devices:** Existing desktop PCs, laptops, or smartphones already present in MeitY/NIC offices (Zero hardware procurement needed).
- **Network:** Standard intranet/web HTTP/HTTPS/WebSocket connectivity (Ports 443 / 8443).

Administrative & Personnel Provisions:

- **Official Identity & Access Passes:** Issuance of MeitY/NIC Technical Consultant Identity Passes for seamless security access to MeitY HQ (CGO Complex, New Delhi) and NIC Regional Center (Salt Lake Sector V, Kolkata).
- **Associate Developer / Contractor Authority:** The Lead Systems Architect (Supratim Dhara) is authorized to onboard and direct up to two (2) associate systems engineers / developers under strict Non-Disclosure Agreements (NDA) to assist with client node deployment, custom WebGPU shader optimization, telemetry monitoring, and benchmark auditing during the pilot duration.
- **Logistics & Travel Support:** Domestic travel allowance covering flight/cab transfers between Delhi HQ and Kolkata regional centers during deployment phases.

- **1x Technical Liaison:** Designated MeitY/NIC engineer to observe packet audits and verify security compliance checkpoints.

6. PILOT DEPLOYMENT FACILITIES & LOCATIONS (DELHI & KOLKATA NODES)

Dual-Region Sovereign Mesh Locations:

- **Node A — MeitY Headquarters (CGO Complex, New Delhi):** Primary FedAvg Signaling Coordinator hosted on 1x NIC cloud Linux server instance. Test workstations in MeitY HQ perimeter.
- **Node B — NIC Regional Center & Webel IT Hub (Salt Lake Sector V, Kolkata):** Edge device cluster utilizing existing idle government office PCs, workstations, and local testbed hardware in West Bengal regional infrastructure.

7. COMMERCIAL STRUCTURE & FINANCIAL BUDGET (PHASE-1 R&D GRANT)

The proposed budget of ₹ 18,00,000 is structured under the **MeitY Sovereign AI R&D Grant / Innovation POC Scheme** pursuant to **General Financial Rules (GFR 2017) Rule 194** for single-source technical innovation procurement:

BUDGET CATEGORY	DESCRIPTION & DELIVERABLES	ALLOCATED AMOUNT (INR)
Systems Architecture & Engineering Retainer	Custom WebGPU/WASM shader compilation, OPFS local storage integration, and FedAvg coordinator protocol customization (Lead Architect Fee & Associate Engineering Support).	₹ 12,50,000
Security Audit & Cryptographic Verification	Packet inspection, Differential Privacy ($\epsilon=0.8$) verification dumps, and Int8 quantization compliance testing under DPDP Act 2023.	₹ 3,50,000
Infrastructure Setup & Technical Operations	NIC Sandbox VM signaling deployment, WebSockets firewall configuration across Delhi & Kolkata nodes, and automated benchmark reporting tools.	₹ 2,00,000
TOTAL PHASE-1 PILOT BUDGET:		₹ 18,00,000 GST Excluded

8. POST-PILOT 90-DAY NATIONAL PRODUCTION SCALING ROADMAP

Following successful sign-off on the Phase-1 Sandbox Pilot, FlockML will transition into a 90-day national scale-out across defense, public infrastructure, and regional government networks:

Month 1 (Days 1–30): Regional Node Expansion & Defense Swarm Trial

- **Node Scaling:** Expand the active mesh grid from 50 sandbox nodes to **1,000+ edge devices** across NIC Delhi HQ, NIC Kolkata (Sector V), Webel IT Parks, and regional state data centers.

- **Defense Integration (iDEX / DRDO):** Initiate a specialized technical trial for collaborative drone autonomy over WebRTC DataChannels, testing un-jammable peer mesh re-sharding when central radio links are lost.

Month 2 (Days 31–60): DPI Integration & Bhashini Sovereign Language Models

- **1-Line SDK Deployment:** Embed `flockml-sdk.js` into high-traffic Digital Public Infrastructure (DPI) portals including DigiLocker, Umang, and National Portal of India.
- **Bhashini Crowdsourced Fine-Tuning:** Utilize idle citizen browser compute cycles to fine-tune regional Indic language models locally, expanding AI accessibility across 22 official Indian languages at \$0.00 hosting cost.

Month 3 (Days 61–90): Orbital Mesh Trial & Permanent Directorate Establishment

- **ISRO Space-Tech Satellite Simulation:** Conduct orbital edge compute simulation evaluating satellite-to-satellite tensor sharing to reduce ground-station bandwidth (beaming 1-KB alert vectors vs 50 GB raw video).
- **Byzantine Fault Tolerance (BFT):** Deploy automated cosine-similarity gradient pruning to secure the global network against malicious data poisoning across 10,000+ active nodes.
- **Permanent Sovereign AI Directorate & Retainer:** Establish the permanent MeitY Sovereign AI Operations Unit, with Supratim Dhara serving as Principal Technical Advisor & Chief Architect overseeing national edge compute infrastructure.

Respectfully Submitted,

Supratim Dhara

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